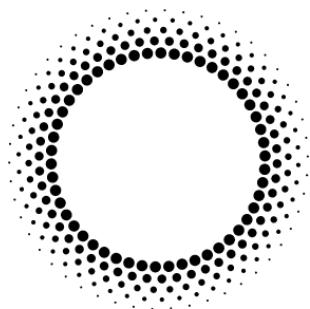




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ML applications

Agentic Context Engineering (ACE): Self-Improving LLMs via Evolving Contexts, Not Fine-Tuning



Editorial Team

October 10, 2025 4 Min Read

(ACE) Self-improving LLMs via evolving contexts, not fine-tuning.

On the Appworld leaderboard on appworld specific tasks (59.74%) IBM QM Q122)
vs CUGA (PUGA (GPT.47) while deepeek v3.1



Reflector



Generator

+10.6%

+86%



Curator

on finance reasoning.

-86.9%
average latency
reduction.

TL;DR: A team of researchers from Stanford University, SambaNova Systems and UC Berkeley introduce [ACE framework](#) that improves LLM performance by editing and growing the input context instead of

Method: Generator → Reflector → Curator

- **Generator** executes tasks and produces trajectories (reasoning/tool calls), exposing helpful vs harmful moves.
- **Reflector** distills concrete lessons from those traces.
- **Curator** converts lessons into typed **delta items** (with helpful/harmful counters) and merges them deterministically, with de-duplication and pruning to keep the playbook targeted.

Two design choices—**incremental delta updates** and **grow-and-refine**—preserve useful history and prevent “context collapse” from monolithic rewrites. To isolate context effects, the research team fixes **the same base LLM (non-thinking DeepSeek-V3.1)** across all three roles.

Benchmarks

AppWorld (agents): Built on the official ReAct baseline, **ReAct+ACE** outperforms strong baselines (ICL, GEPA, Dynamic Cheatsheet), with **+10.6% average** over selected baselines and **~+7.6%** over Dynamic Cheatsheet in online adaptation. On the **Sept 20, 2025** leaderboard, **ReAct+ACE 59.4% vs IBM CUGA 60.3% (GPT-4.1)**; ACE surpasses CUGA on the harder **test-challenge** split, while using a smaller open-source base model.

Finance (XBRL): On **FiNER** token tagging and **XBRL Formula** numerical reasoning, ACE reports **+8.6% average** over baselines with ground-truth labels for offline adaptation; it also works with execution-only feedback, though quality of signals matters.

Method	GT Labels	Test-Normal		Test-Challenge		Average
		TGC↑	SGC↑	TGC↑	SGC↑	
DeepSeek-V3.1 as Base LLM						
ReAct		63.7	42.9	41.5	21.6	42.4
Offline Adaptation						
ReAct + ICL	✓	64.3 _{+0.6}	46.4 _{+3.5}	46.0 _{+4.5}	27.3 _{+5.7}	46.0 _{+3.6}
ReAct + GEPA	✓	64.9 _{+1.2}	44.6 _{+1.7}	46.0 _{+4.5}	30.2 _{+8.6}	46.4 _{+4.0}
ReAct + ACE	✓	76.2 _{+12.5}	64.3 _{+21.4}	57.3 _{+15.8}	39.6 _{+18.0}	59.4 _{+17.0}
ReAct + ACE	✗	75.0 _{+11.3}	64.3 _{+21.4}	54.4 _{+12.9}	35.2 _{+13.6}	57.2 _{+14.8}
Online Adaptation						
ReAct + DC (CU)	✗	65.5 _{+1.8}	58.9 _{+16.0}	52.3 _{+10.8}	30.8 _{+9.2}	51.9 _{+9.5}
ReAct + ACE	✗	69.6 _{+5.9}	53.6 _{+10.7}	66.0 _{+24.5}	48.9 _{+27.3}	59.5 _{+17.1}

Table 1: **Results on the AppWorld Agent Benchmark.** "GT labels" indicates whether ground-truth labels are available to the Reflector during adaptation. We evaluate the ACE framework against multiple baselines on top of the official ReAct implementation, both for offline and online context adaptation. ReAct + ACE outperforms selected baselines by an average of 10.6%, and could achieve good performance even without access to GT labels.

Method	GT Labels	Test-Normal		Test-Challenge		Average
		TGC↑	SGC↑	TGC↑	SGC↑	
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Offline Adaptation						
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ReAct + GEPA	✓	64.9 _{+1.2}	44.6 _{+1.7}	46.0 _{+4.5}	30.2 _{+8.6}	46.4 _{+4.0}
ReAct + ACE	✓	76.2 _{+12.5}	64.3 _{+21.4}	57.3 _{+15.8}	39.6 _{+18.0}	59.4 _{+17.0}
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Method	GT Labels	FINER (Acc↑)	Formula (Acc↑)	Average
DeepSeek-V3.1 as Base LLM				
Base LLM		70.7	67.5	69.1
Offline Adaptation				
ICL	✓	72.3 ^{+1.6}	67.0 ^{−0.5}	69.6 ^{+0.5}
MIPROv2	✓	72.4 ^{+1.7}	69.5 ^{+2.0}	70.9 ^{+1.8}
GEPA	✓	73.5 ^{+2.8}	71.5 ^{+4.0}	72.5 ^{+3.4}
ACE	✓	78.3 ^{+7.6}	85.5 ^{+18.0}	81.9 ^{+12.8}
ACE	✗	71.1 ^{+0.4}	83.0 ^{+15.5}	77.1 ^{+8.0}
Online Adaptation				
DC (CU)	✓	74.2 ^{+3.5}	69.5 ^{+2.0}	71.8 ^{+2.7}
DC (CU)	✗	68.3 ^{−2.4}	62.5 ^{−5.0}	65.4 ^{−3.7}
ACE	✓	76.7 ^{+6.0}	76.5 ^{+9.0}	76.6 ^{+7.5}
ACE	✗	67.3 ^{−3.4}	78.5 ^{+11.0}	72.9 ^{+3.8}

Table 2: **Results on Financial Analysis Benchmark.** "GT labels" indicates whether ground-truth labels are available to the Reflector during adaptation. With GT labels, ACE outperforms selected baselines by an average of 8.6%, highlighting the advantage of structured and evolving contexts for domain-specific reasoning. However, we also observe that in the absence of reliable feedback signals (*e.g.*, ground-truth labels or execution outcomes), both ACE and other adaptive methods such as Dynamic Cheatsheet may degrade, suggesting that context adaptation depends critically on feedback quality.

Method	GT Labels	FINER (Acc↑)	Formula (Acc↑)	Average
DeepSeek-V3.1 as Base LLM				
Base LLM		70.7	67.5	69.1
Offline Adaptation				
ICL	✓	72.3 ^{+1.6}	67.0 ^{−0.5}	69.6 ^{+0.5}
MIPROv2	✓	72.4 ^{+1.7}	69.5 ^{+2.0}	70.9 ^{+1.8}
GEPA	✓	73.5 ^{+2.8}	71.5 ^{+4.0}	72.5 ^{+3.4}
ACE	✓	78.3 ^{+7.6}	85.5 ^{+18.0}	81.9 ^{+12.8}
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Online Adaptation				
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DC (CU)	✗	68.3 ^{−2.4}	62.5 ^{−5.0}	65.4 ^{−3.7}
ACE	✓	76.7 ^{+6.0}	76.5 ^{+9.0}	76.6 ^{+7.5}
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Table 2: **Results on Financial Analysis Benchmark.** "GT labels" indicates whether ground-truth labels are available to the Reflector during adaptation. With GT labels, ACE outperforms selected baselines by an average of 8.6%, highlighting the advantage of structured and evolving contexts for domain-specific reasoning. However, we also observe that in the absence of reliable feedback signals (*e.g.*, ground-truth labels or execution outcomes), both ACE and other adaptive methods such as Dynamic Cheatsheet may degrade, suggesting that context adaptation depends critically on feedback quality.

ACE's non-LLM merges plus localized updates reduce adaptation overhead substantially:

- Offline (AppWorld): -82.3% latency and -75.1% rollouts vs GEPA.
- Online (FiNER): -91.5% latency and -83.6% token cost vs Dynamic Cheatsheet.

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		TGC↑	SGC↑	TGC↑	SGC↑	
DeepSeek-V3.1 as Base LLM						
ReAct		63.7	42.9	41.5	21.6	42.4
Offline Adaptation						
ReAct + ACE w/o Reflector or multi-epoch	✓	70.8 ^{+7.1}	55.4 ^{+12.5}	55.9 ^{+14.4}	38.1 ^{+17.5}	55.1 ^{+12.7}
ReAct + ACE w/o multi-epoch	✓	72.0 ^{+8.3}	60.7 ^{+17.8}	54.9 ^{+13.4}	39.6 ^{+18.0}	56.8 ^{+14.4}
ReAct + ACE	✓	76.2 ^{+12.5}	64.3 ^{+21.4}	57.3 ^{+15.8}	39.6 ^{+18.0}	59.4 ^{+17.0}
Online Adaptation						
ReAct + ACE	✗	67.9 ^{+4.2}	51.8 ^{+8.9}	61.4 ^{+19.9}	43.2 ^{+21.6}	56.1 ^{+13.7}
ReAct + ACE + offline warmup	✗	69.6 ^{+5.9}	53.6 ^{+10.7}	66.0 ^{+24.5}	48.9 ^{+27.3}	59.5 ^{+17.1}

Table 3: **Ablation Studies on AppWorld.** We study how particular design choices of ACE (iterative refinement, multi-epoch adaptation, and offline warmup) could help high-quality context adaptation.

Method	Latency (s)↓	# Rollouts↓	Method	Latency (s)↓	Token Cost (\$)↓
ReAct + GEPA	53898	1434	DC (CU)	65104	17.7
ReAct + ACE	9517 ^(-82.3%)	357 ^(-75.1%)	ACE	5503 ^(-91.5%)	2.9 ^(-83.6%)

(a) **Offline** (AppWorld). (b) **Online** (FiNER).

Table 4: **Cost and Speed Analysis.** We measure the context adaptation latency, number of rollouts, and dollar costs of ACE against GEPA (offline) and DC (online).

Method	GT Labels	Test-Normal		Test-Challenge		Average
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DeepSeek-V3.1 as Base LLM						
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(a) **Offline** (AppWorld). (b) **Online** (FiNER).

Table 4: **Cost and Speed Analysis.** We measure the context adaptation latency, number of rollouts, and dollar costs of ACE against GEPA (offline) and DC (online).

- **ACE = context-first adaptation:** Improves LLMs by incrementally editing an evolving “playbook” (delta items) curated by Generator→Reflector→Curator, using the **same** base LLM (non-thinking DeepSeek-V3.1) to isolate context effects and avoid collapse from monolithic rewrites.
- **Measured gains:** ReAct+ACE reports +10.6% over strong baselines on AppWorld and achieves 59.4% vs IBM CUGA 60.3% (GPT-4.1) on the Sept 20, 2025 leaderboard snapshot; finance benchmarks (FiNER + XBRL Formula) show +8.6% average over baselines.
- **Lower overhead than reflective-rewrite baselines:** ACE reduces adaptation latency by ~82–92% and rollouts/token cost by ~75–84%, contrasting with Dynamic Cheatsheet’s persistent memory and GEPA’s Pareto prompt evolution approaches.

Conclusion

ACE positions context engineering as a first-class alternative to weight updates: maintain a persistent, curated playbook that accumulates task-specific tactics, yielding measurable gains on AppWorld and finance reasoning while cutting adaptation latency and token rollouts versus reflective-rewrite baselines. The approach is practical—deterministic merges, delta items, and long-context-aware serving—and its limits are clear: outcomes track feedback quality and task complexity. If adopted, agent stacks may “self-tune” primarily through evolving context rather than new checkpoints.

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Asif Razzaq is the CEO of Marktechpost Media Inc.. As a visionary entrepreneur and engineer, Asif is committed to harnessing the potential of Artificial Intelligence for social good. His most recent endeavor is the launch of an Artificial Intelligence Media Platform, Marktechpost, which stands out for its in-depth coverage of machine learning and deep learning news that is both technically sound and easily understandable by a wide audience. The platform boasts of over 2 million monthly views, illustrating its popularity among audiences.



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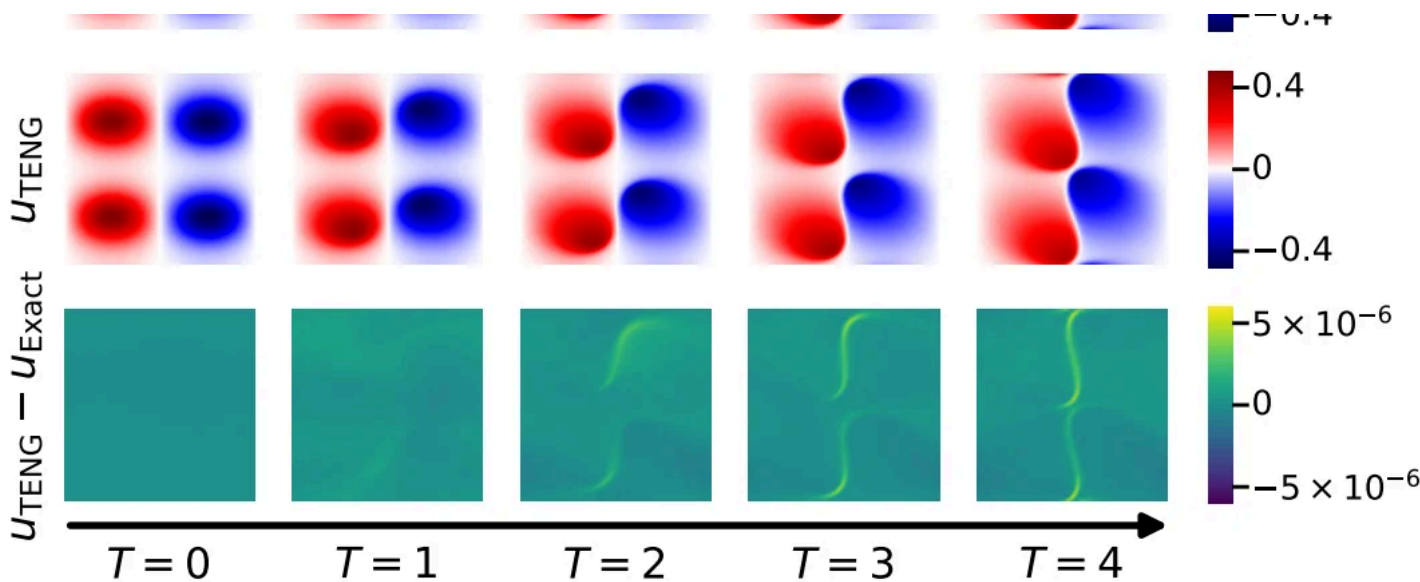


Figure 3. Reference solution, TENG solution, and the difference

Transforming Partial Differential Equations PDE Solutions with 'TENG': Harnessing Machine Learning for Enhanced Accuracy and Efficiency

Partial differential equations (PDEs) are required for modeling dynamic systems in science and engineering,

By Editorial Team April 22, 2024

id 38 k to 45 k salary expectation
ing interview , was offered 38 k . Have
leverage , and overall its a great
ry level position and company
ould I negotiate ?

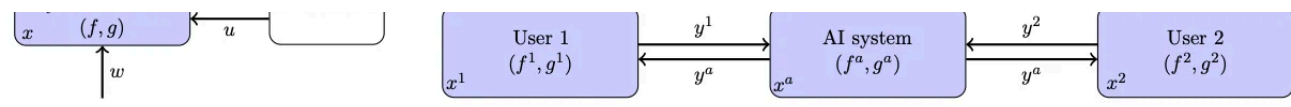
S aid 38 k to 45 k salary expectation
during interview , was offered 250 k . H
no leverage , and overall its a great
management position and company
. Should I negotiate ?

Figure 1: Credit assignment in DPO based on answer-level feedback. We provide two summaries of a Reddit post about a job interview. The left is the base response and on the right we provide a response with reduced errors in the salary range and the position level. Each token is colored corresponding to the DPO implicit reward as expressed in Eq. 11 (darker is higher), using the trained model. We observe that the model correctly highlights the erroneous statements, without much change to the value of the other tokens, which indicates the ability to do credit assignment.

Researchers at Stanford University Explore Direct Preference Optimization (DPO): A New Frontier in Machine Learning and Human Feedback

Exploring the synergy between reinforcement learning (RL) and large language models (LLMs) reveals a

By Editorial Team April 22, 2024



(a) Model and environment interaction (b) Formal interaction model as a coupled dynamical system

Figure 1: Left: illustration of a dynamical system where a model interacts with an unspecified environment. Right: illustration of a formal interaction model as a coupled dynamical system between an AI system and users. The variable x captures the internal state, the function f captures the state update, the function g captures the measurement equation, the variable u denotes inputs, the variable y denotes outputs (and inputs, in the case of the coupled dynamical system), and the variable w captures noise or other disturbance (Section 2.1).

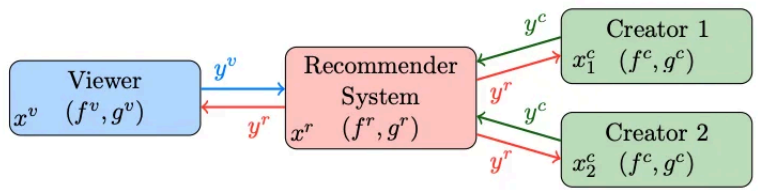


Figure 2: Illustration of a formal interaction model for a content recommendation system as a coupled dynamical system between the recommender system (x^r, x^r, f^r, g^r) , viewers (x^v, x^v, f^v, g^v)

Formal Interaction Model (FIM): A Mathematics-based Machine Learning Model that Formalizes How AI and Users Shape One

Another

Machine learning has become an important domain that has contributed to developing platforms and

By Editorial Team April 22, 2024



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